

Answer Key — Biology Grade 11

Sub-Strand 2.3: Excretion in Plants • Lesson 5: Who Profits from Plant Wastes? Community Uses, Alkaloid Risks and the Final Explanation

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	B	C	D	C	D	D

USE AFTER: PREDICT PHASE — Predict: six labelled stations and the question 'could any of them harm people?'

Q1. Tea leaves from Kericho contain caffeine and miraa (khat) contains cathinone. Both are alkaloids. Which statement about alkaloids is correct?

- A. Some alkaloids are useful in small controlled doses but harmful when abused**
- B. All alkaloids are harmful and should never be used
- C. Alkaloids are harmless and cannot cause addiction
- D. Only alkaloids from tobacco and miraa can harm people

Correct answer: A. Alkaloids such as caffeine and quinine are useful in small doses, but misuse (for example of nicotine or miraa) leads to addiction and health problems. The dose and the abuse make the difference.

USE AFTER: OBSERVE PHASE — Observe: hand lens re-examination of the hardened Acacia gum lump and dried Euphorbia latex sample

Q2. A farmer cuts the stem of a Euphorbia hedge and thick milky latex oozes out and later hardens. Which excretory process released the latex?

- A. Organ fall
- B. Exudation**
- C. Transpiration
- D. Guttation

Correct answer: B. Exudation is the release of products such as latex, gum and resin from plant tissues, especially from cuts. Guttation releases water droplets, transpiration loses water vapour and organ fall removes leaves.

USE AFTER: OBSERVE PHASE — Observe: portfolio presentation and gallery rotation (tannins in Acacia/wattle bark used for tanning hides in Nakuru)

Q3. In Nakuru, tanners soak animal hides in a solution made from the bark of Acacia or wattle trees. Which plant excretory product in the bark makes this possible, and what is it used for?

- A. Pyrethrum, used to make insecticide
- B. Quinine, used to treat malaria
- C. Tannins, used for tanning hides into leather**
- D. Caffeine, used to flavour drinks

Correct answer: C. Tannins stored in Acacia and wattle bark are used to tan hides. Caffeine is in tea and coffee, pyrethrum is an insecticide and quinine is a malaria medicine.

USE AFTER: EXPLAIN PHASE — Explain: CER matrix evidence (guttation droplets at the edges of arrowroot leaves at 6:30 am) and the Daily Exit Ticket

Q4. Early in the morning, droplets of liquid water are seen along the edges of arrowroot (taro) leaves. Which statement correctly contrasts guttation with transpiration?

- A. Guttation loses water vapour through stomata, while transpiration loses liquid droplets through hydathodes
- B. Guttation and transpiration are the same process and both happen only at night
- C. Guttation removes sticky resin from cut stems, while transpiration removes fallen leaves
- D. Guttation loses liquid water droplets through hydathodes at leaf edges, while transpiration loses water vapour mainly through stomata**

Correct answer: D. Guttation is the loss of liquid water from hydathodes, often in humid early mornings. Transpiration is the loss of water vapour, mainly through stomata.

USE AFTER: EXPLAIN PHASE — Explain: CER matrix reasoning column (why plants lack excretory organs)

Q5. A learner asks, 'Why does a plant not need a kidney?' Which answer is the best scientific reasoning?

- A. Plants produce very toxic wastes that must be kept inside the cells at all times
- B. Plants absorb all their wastes from the soil and release them through the roots
- C. Plants have low metabolic rates and less toxic wastes, recycle CO₂ and O₂ between photosynthesis and respiration, and store other wastes in dead tissues**
- D. Plants do not produce any wastes

Correct answer: C. Plants make less toxic wastes, recycle gases between photosynthesis and respiration, and store wastes such as tannins and resins in dead tissues or lose them by organ fall, so they need no kidneys.

USE AFTER: EXPLAIN PHASE — Explain: CER matrix evidence column (interview data, 7 of 12 households use neem)

Q6. A group interviewed 12 households in their community. 7 of the 12 households said they use neem leaves to treat malaria or skin problems. What percentage of the households use neem?

- A. 7%
- B. 70%
- C. 41.7%
- D. 58.3%**

Correct answer: D. Percentage = $7 \div 12 \times 100 = 58.3\%$. The value 41.7% is the share that do not use neem (5 out of 12).

USE AFTER: MODEL BUILDING PHASE — Model: comparing the Lesson 1 model with the Final Model using blue and red pens

Q7. In Lesson 1, a learner drew a model showing that plants have no wastes. After Lessons 1-5, which is the correct revised idea for the final model?

- A. Plants have no wastes, so nothing needs to be removed
- B. Plants produce wastes, but these have no use to people
- C. Plants produce wastes, but all of them are removed through kidneys in the roots
- D. Plants produce wastes, but they remove or store them in different ways, such as exudation, guttation, deposition and organ fall**

Correct answer: D. The evidence shows that plants do produce wastes. They remove them by diffusion, transpiration, guttation and exudation, or store them in tissues and lose them by organ fall, and many have economic value.